

5. Build and Verify

A weekend to build, 48 hours to cure

Measure, inspect, remake, prove

THIS IS A PROJECTILE WEAPON

A combustion launcher throws a half-pound potato several hundred feet at speeds that will kill a person. It is not a toy, it is not a firework, and the fact that it is made of plumbing parts changes nothing about what it does downrange. Every rule on the **Safety** sheet applies every time: adult present and in charge, muzzle discipline as strict as a rifle's, a confirmed clear range, eyes and ears on, and never a person, animal, building, road or anything you cannot afford to destroy in the direction it is pointing.

The standard

Do not reward a finished-looking launcher. Reward a documented sequence of square cuts, clean sockets, fully seated joints, a complete cure, and an adult's final decision. A doubtful pressure part is not repaired or "good enough": mark it REJECT, make it unusable, and replace it.

Parts and acceptance

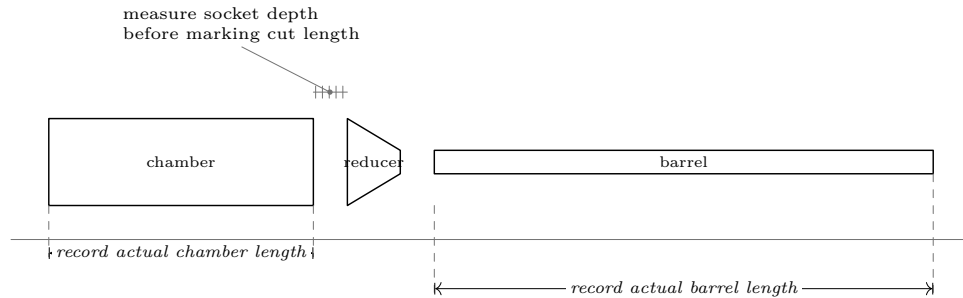
Part	Required evidence	Reject if
Chamber	3 in Sch 40 pressure PVC; rating and manufacturer print legible	DWV, foam/cellular core, unknown rating, crack, whitening, deep gouge
Cleanout	Matching pressure-rated adapter and undamaged threaded plug	Plug binds, rocks, or has damaged threads; plug must never be glued
Reducer	Pressure-rated fitting; sockets clean and round	Socket chipped, distorted, contaminated, or a loose dry fit
Barrel	1.5 in Sch 40 pressure PVC; straight and fully printed	Split, deep score, distortion, or unknown rating
Igniter assembly	Piezo unit; stainless screws, nuts, and washers	Loose hardware, damaged insulation, unreliable spark
Joining materials	Compatible primer and pressure-PVC cement, in date	Gel, skin, contamination, missing label, expired shelf life

Pressure-rated pipe only. Schedule 40 *pressure* PVC, printed with an NSF-PW rating and a pressure figure. Never DWV, never foam-core, never cellular-core, never thin-wall drain pipe — those are not rated for internal pressure at all and they fail by shattering into fragments rather than splitting. This is the one specification in the whole project that is not negotiable.

Before touching a tool

Question	Record	Adult check
Can you point to the pressure rating on every pressure part?	Rating: _____	Initials/date: _____
Which face will be the chamber end of each fitting?	Pencil arrows made: yes / no	All mating faces identified: yes / no
Where will the assembly cure untouched for 48 hours?	Place: _____	Start time: _____

1. Lay out and measure



1. Place every dry, uncemented part in order. Match the printed ratings to the parts list. Photograph or sketch the rating marks before they disappear under fittings.
2. Measure the depth of each socket with a rule. Mark insertion depth on each pipe end, then draw an alignment line across each future joint.
3. Measure the pipe twice at points 90 degrees apart. If the readings differ, stop: the end is out of round.
4. Dry-fit without forcing. Every pipe must reach its depth mark. Never “fix” a loose fit with extra cement.

Layout record

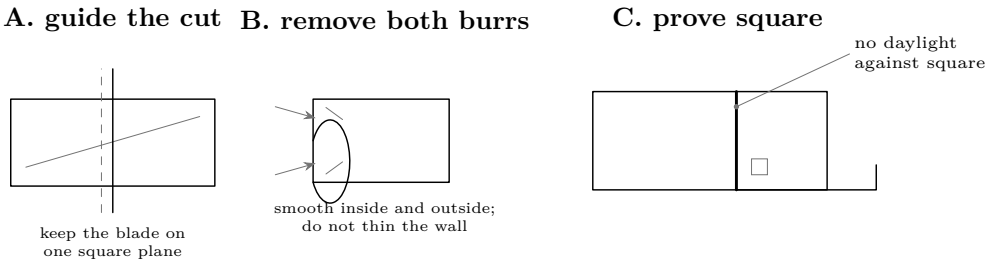
Feature	Planned	Measured	Pass / remake
Chamber cut length	_____	_____	_____
Barrel cut length	_____	_____	_____
Reducer socket depth	_____	_____	_____
Cleanout socket depth	_____	_____	_____

Pause and explain. Why does a socket-depth mark reveal an incomplete joint? Why is a square end stronger than a slanted end? Have the builder answer before cutting. *see the **01 Measure and Mark** sheet*

Rating-mark evidence

Part	Mark copied or photo ID	Witness / decision
Chamber		
Barrel		
Reducer and cleanout		

2. Cut, deburr, and inspect



1. Clamp the pipe without crushing it. Cut at the waste side of the mark, using a mitre box or pipe cutter. Do not hurry the final stroke.
2. Deburr the full circumference inside and out. Make only a small chamfer; the pressure wall must remain full thickness.
3. Hold a square at four positions around the end. Shine a light behind it. Record the largest visible gap with a feeler gauge or rule.
4. Wipe the end clean with a dry white cloth. Inspect under raking light.

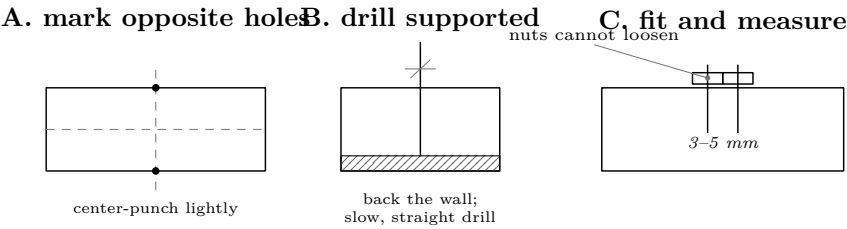
End	Gap	Inner burr	Outer burr	Decision / initials
Chamber A	_____	none / found	none / found	_____
Chamber B	_____	none / found	none / found	_____
Barrel A	_____	none / found	none / found	_____
Barrel B	_____	none / found	none / found	_____

REMAKE DECISION

If an end is visibly slanted, chipped, cracked, deeply scored, heat-damaged, or too short after correction, do not advance it. Mark the defect, cut back to sound material if the planned dimensions still permit, or replace the pipe. Never fill a bad cut with cement or epoxy.

Raking-light evidence		
End	Observation and largest gap	Accept / remake; initials
Chamber ends		
Barrel ends		

3. Drill and prove the ignition assembly



1. Mark hole centers on the agreed chamber location. Confirm that hardware will clear the cleanout, reducer, and hands.

2. Support the inside of the pipe. Drill small pilot holes, then only to the diameter required by the fasteners. Deburr without enlarging them.

3. Install washers and nuts. Set the internal electrode gap to 3–5 mm. Seal only as specified on the craft sheet; sealant does not compensate for a loose or oversized hole.

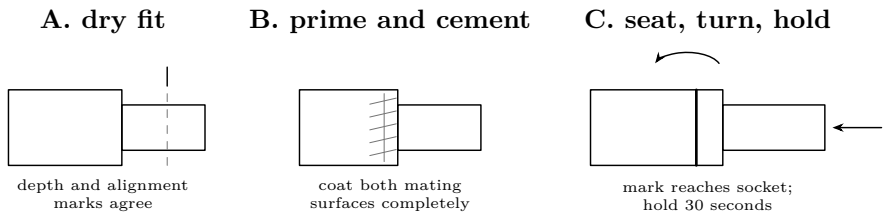
4. With the cleanout open, chamber empty, and no fuel present, have the adult operate the piezo. Observe from the open end, never over the chamber.

Ignition inspection		
Check	Measurement / result	Decision
Electrode gap at closest point	_____ mm	pass / remake
Ten empty-chamber clicks	_____ sparks seen	pass / remake
Hardware movement by hand	none / present	pass / remake
Hole edge under raking light	clean / cracked / enlarged	pass / remake

Pause and explain. Why test the ignition before cementing the chamber? What defect would make replacement safer than adding more sealant? *see the 05 Drill and Seal sheet*

Ten-click witness record		
Clicks	Observed spark count / pattern	Adult witness / decision
1–5		
6–10		

4. Dry-fit, solvent-weld, and witness



1. Rehearse the complete movement dry: orient, insert fully, quarter-turn, hold. Assign one adult to read time and one to witness the marks.
2. Work one joint at a time in ventilation specified by the product labels. Prime both mating surfaces, then cement both; follow the manufacturer’s application and temperature instructions.
3. Push to the socket-depth mark, turn a quarter turn, restore the alignment mark, and hold for 30 seconds. Do not disturb the joint.
4. Wipe only the exterior excess. Never scrape, heat, pressure-test, or “exercise” a fresh joint.

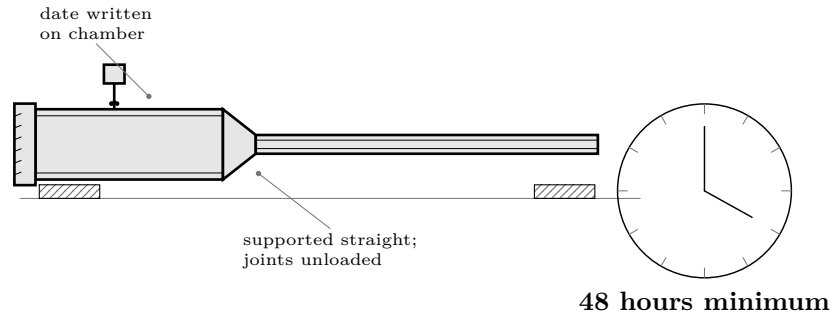
Joint	Start time	Held to	Mark seated	Witness / decision
Cleanout–chamber	_____	_____	yes / no	_____
Chamber–reducer	_____	_____	yes / no	_____
Reducer–barrel	_____	_____	yes / no	_____

STOP, DO NOT PATCH

Reject a joint that backs out, misses its depth mark, rotates away from its alignment mark, contains a visible dry area, or was assembled after the cement had skinned. Do not add cement around the outside. Retire the affected pressure part and remake the assembly. *see the 04 Solvent Weld sheet*

Fresh-joint witness		
Joint	Full-rotation bead observation	Witness / disposition
Cleanout–chamber		
Chamber–reducer		
Reducer–barrel		

5. Cure and inspect



1. Write assembly date and time on the chamber. Place it horizontally on two broad supports where nobody will lift, twist, load, or test it.
2. Leave it at least 48 hours, and longer if the cement label requires it for the actual pipe size, temperature, or humidity. The label governs.
3. After cure, inspect every joint in bright raking light. Turn the assembly slowly. Look for a continuous seated shoulder, alignment, cracks, whitening, bulges, gaps, and disturbed witness marks.
4. Inspect the cleanout threads dry. The plug must run fully home by hand without cross-threading. Never glue, tape over, or force damaged threads.

Cure record

Evidence	Record	Adult initials
Assembly finished	date/time: _____	_____
Earliest allowed inspection	date/time: _____	_____
Actual inspection	date/time: _____	_____
Label required longer cure?	no / yes: _____	_____

360-degree inspection

Feature	Result	Defect location / decision
Cleanout joint and threads	pass / reject	_____
Chamber surface and electrodes	pass / reject	_____
Reducer joints	pass / reject	_____
Barrel surface and muzzle	pass / reject	_____

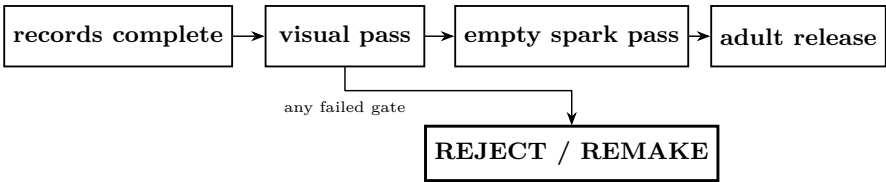
Cure arithmetic. Label minimum _____ hours; project minimum 48 hours.

Later governing time: _____ Actual elapsed time: _____ Checked by: _____

Inspection notes

Area	Lighting, angle, and observation	Decision
Threads and plug		
Joints and wall		

6. Final proof: permission to proceed



Gate	Result	Evidence
Every pressure part was identified and accepted before work	pass / reject	Parts record complete
Every cut and joint has a measurement and witness decision	pass / reject	Pages 2–5 complete
At least 48 hours and the label-required cure elapsed	pass / reject	Cure times complete
Full surface and all joints passed bright-light inspection	pass / reject	No defect left blank
Cleanout operates by hand; electrodes are immobile	pass / reject	Adult observed
Ten empty, unfuelled ignition clicks produced ten clear sparks	pass / reject	Count: ____/10
Safety sheet was read aloud at the chosen legal range	pass / reject	Reader: _____

NO IMPROVISED PRESSURE PROOF

Do not plug the barrel, add compressed air, hydrostatically pressurize this assembly, strike it, heat it, or perform a close-range “test shot.” Those tests introduce hazards and do not convert questionable workmanship into proof. A failed inspection means retire and remake.

Adult release

I reviewed the complete record, inspected the assembly after cure, confirmed that every gate above passes, and accept responsibility for deciding whether and where any later operation is lawful and safe.

Adult name: _____ Signature: _____ Date: _____

Operation is a separate gate

This build record gives no permission to fire. Before any use, the supervising adult must apply the **Safety** sheet, local law, landowner rules, weather, backstop, exclusion zone, eye and hearing protection, muzzle control, and misfire procedure. Use only the single approved operating procedure; do not vary fuel, projectile fit, barrel, angle, or other variables to seek more pressure, speed, or range. If performance seems abnormal, stop and inspect; never “tune” a combustion launcher.

Final question. What evidence would make you retire this launcher today? Write three observations before any operation: _____, _____, and _____.

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